

RESEARCH & PROJECT PORTFOLIO

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Data Center Thermal Management · EMNSPC Lab, The University of Texas at Arlington

Ph.D. expected Dec 2026 · Open to full-time roles

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AT A GLANCE

From lab research to field deployment

Ph.D. candidate at UT Arlington's EMNSPC Lab working on data center cooling: two-phase and single-phase immersion, indirect evaporative cooling, liquid cold plates and thermal interface materials. I combine experimental characterization with CFD, and I've taken that work into the field on a 10 MW deployment.

Ph.D., Mechanical Engineering
UT Arlington · Expected Dec 2026

M.S., Mechanical Engineering
UT Arlington · 2021

B.E., Mechanical Engineering
University of Pune · 2017

Seeking: full-time roles in data center thermal management

10 MW
Two-phase immersion cooling site deployed and commissioned

5+
Industry partners, including LiquidStack, Chemours and MESTEX

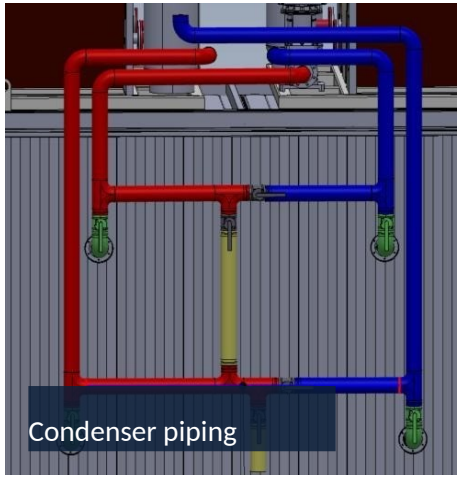
5×
Lower cost: in-house TIM tester vs. commercial units

8
Conference papers and posters (InterPACK, ITherm, ES2)

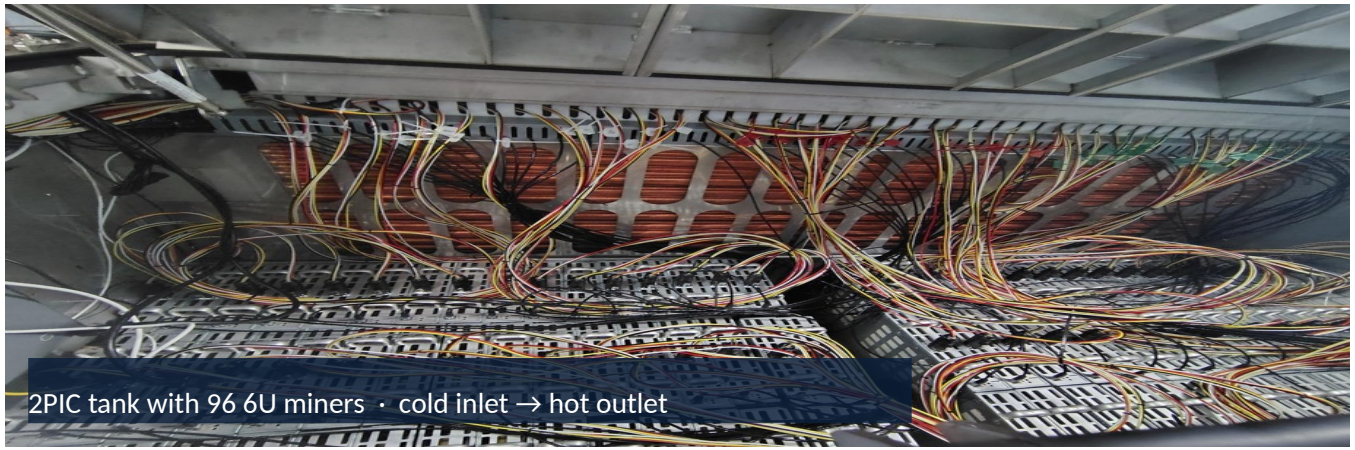
10 MW two-phase immersion cooling deployment



Dry coolers · up to 500 kW each



Condenser piping



2PIC tank with 96 6U miners · cold inlet → hot outlet

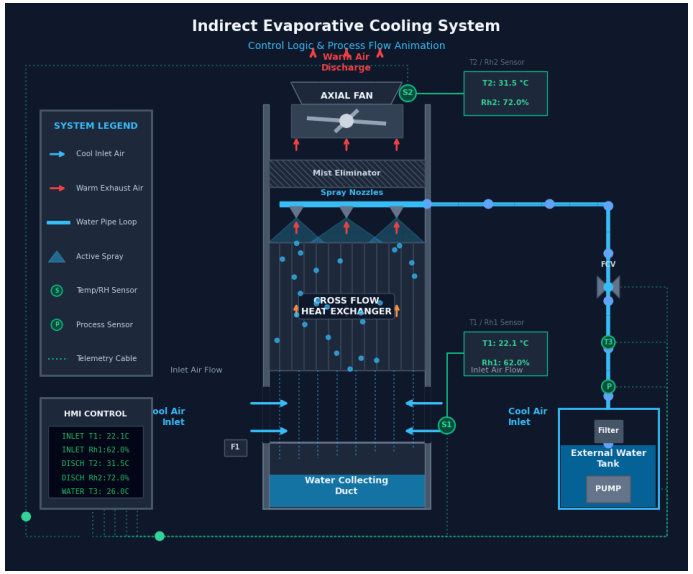
10 MW
Site capacity

1 MW
Per tank

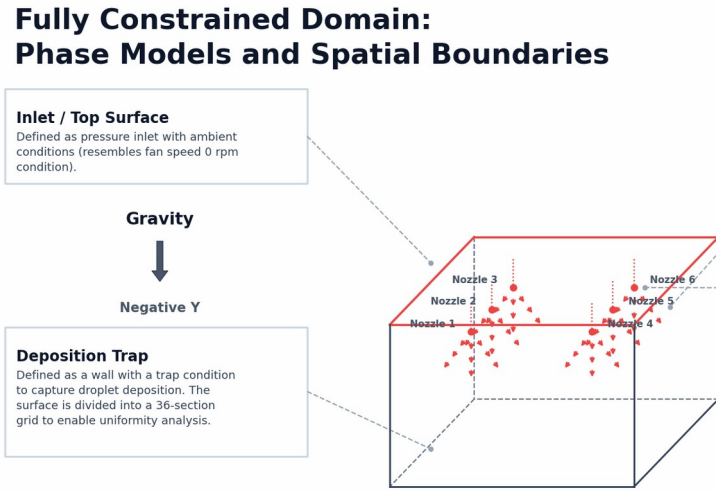
500 kW
Per dry cooler

- Led on-site deployment, installation and commissioning of the two-phase immersion cooling (2PIC) site
- Managed planning, resources and timelines; coordinated MEP contractors and cross-functional teams
- Ran miner energization, SCADA, sensor instrumentation and data acquisition; sensor analysis enabled miner overclocking
- Validated thermal interfaces, fluid levels and pressure integrity; executed leak testing and performance validation

Indirect evaporative cooling for data centers



Experimental setup: control logic and process flow



CFD domain in ANSYS Fluent: six-nozzle spray model

- Characterized IEC performance under varying airflow, water flow rate and nozzle height
- Analyzed heat and mass transfer, pressure drop and spray distribution with psychrometric analysis
- Built and validated a CFD model of six-nozzle spray distribution against experimental heat maps
- Flownex model build and developed

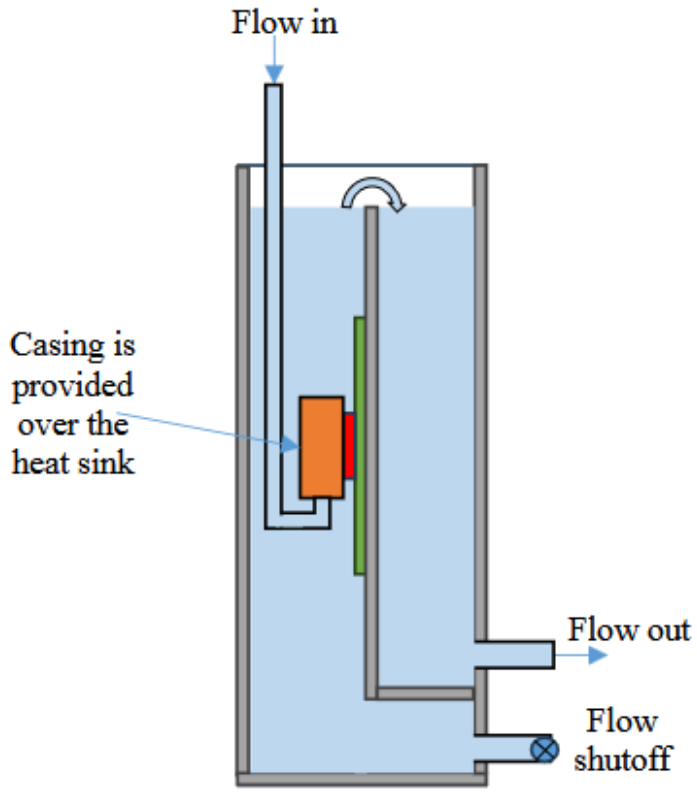
OUTCOMES

Ph.D. research
This project forms the core of my doctoral work

2 journal manuscripts
In preparation

2 posters
InterPACK 2026 · ES2 2026

Single-phase immersion cooling: heat sink optimization



Experimental setup for targeted flow

Test setup

Co-designed and built a single-phase immersion tank with LiquidStack and characterized multiple heat-sink designs

CFD model

Developed a CFD model in ANSYS Icepak and validated it against experimental results

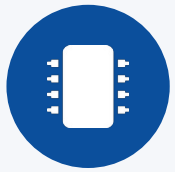
Heat sink optimization

Optimized the design with CFD, then manufactured and tested it to confirm the improvement

Publications

IEEE ITherm 2025 paper on targeted flow in aluminum and copper heat sinks; journal manuscript in preparation

Two-phase immersion: from component to field



STEP 1

Component-level testing

Two-phase immersion experiments on individual components, in collaboration with Chemours



STEP 2

Server-level testing

Scaled the experimental testing up to full servers in two-phase immersion

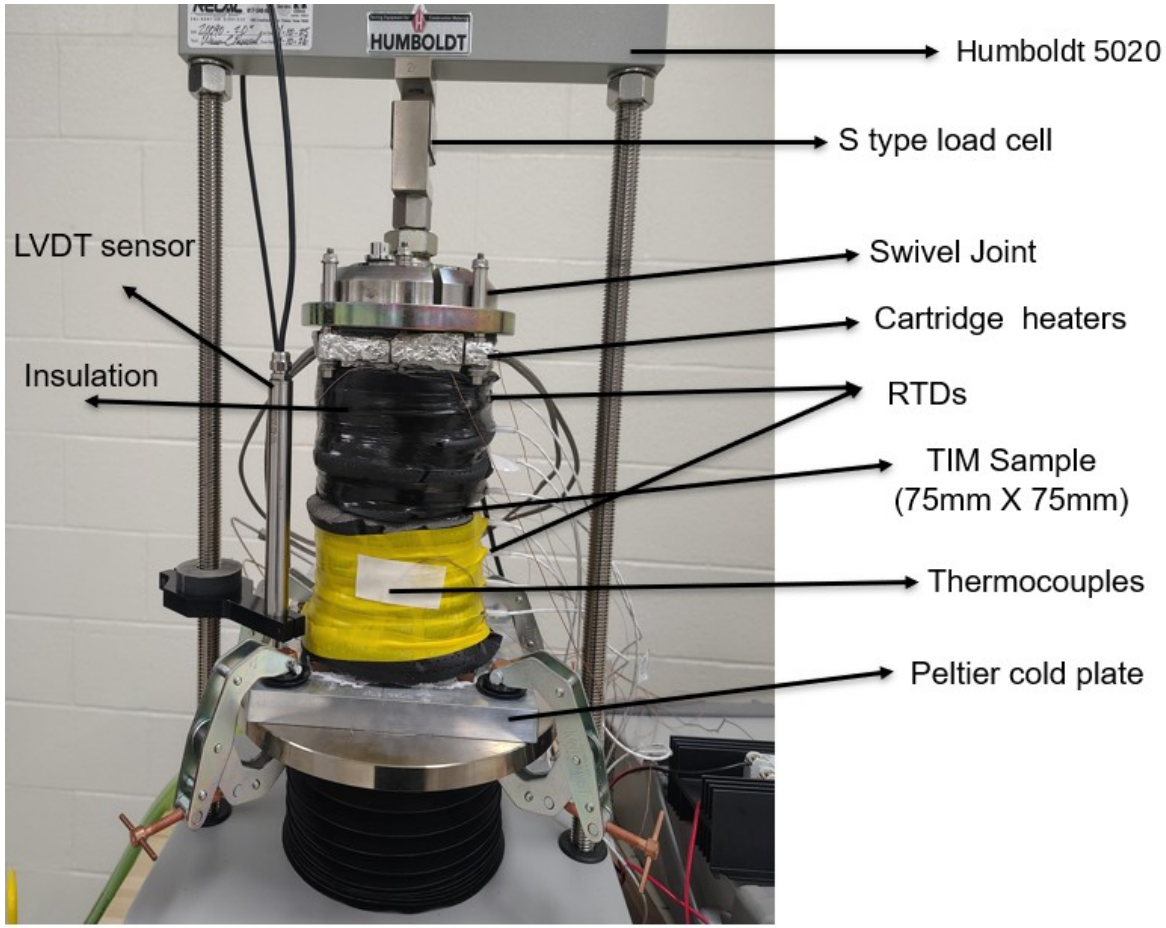


STEP 3

Field deployment

Carried that experience into a 10 MW two-phase immersion site at Marathon Digital Holdings (2025)

Reliability thermal interface material (TIM) tester



5×
Lower cost than commercial testers

3×
Larger test surface area

75×75
mm test surface

- Measures thermal resistance and apparent thermal conductivity under controlled pressure, interface area and material conditions
- Integrates cartridge heaters, copper hot blocks, RTDs, load cells and an LVDT with a compression testing frame
- The larger test area brings results closer to real, large-area thermal management applications

OPEN COMPUTE PROJECT · SHELL, EXXONMOBIL, VALVOLINE

Immersion fluid reliability study

Evaluating long-term material compatibility and fluid reliability for immersion cooling systems

1

Prepare

Prepared material coupons, pre-weighed samples and formulated test fluids

2

Age

Ran controlled aging batches of coupons in the test fluids

3

Post-process

Cleaned, imaged and re-weighed coupons to quantify weight loss

4

Characterize

Analyzed changes in fluid chemistry, material composition and surfaces

ANALYSIS TECHNIQUES

FTIR | Total Acid Number (TAN) | Conductivity | ICP | SEM | XPS

EARLIER WORK · 2019 – 2021

Data center thermal research since 2019



M.S. THESIS · UT ARLINGTON, 2021

Thermal Design Analysis of Server Chassis Manifolds for Liquid-Cooled Servers Using CFD

CFD-based thermal design of the manifolds that distribute coolant inside liquid-cooled server chassis.



ASME InterPACK 2020 · PAPER & POSTER

CFD Modeling of the Distribution of Airborne Particulate Contaminants Inside Data Center Hardware

Co-authored a peer-reviewed paper and presented a technical poster on airborne particulate behavior in data center environments.

SKILLS & CERTIFICATIONS

Tools and techniques

Thermal testing & measurement

Thermocouples, RTDs, Agilent DAQ, environmental chambers, CDUs (L2L/L2A), thermal test vehicles, cold plates, Belimo valves, chillers, dry coolers, flow & pressure sensors

CAD & CAE

SolidWorks, AutoCAD, ANSYS Workbench, ANSYS Fluent, ANSYS Icepak, Flownex

Controls, data & software

MATLAB, SCADA, data acquisition, Microsoft Office, project planning tools

Testing & operations

Test plan development, root-cause analysis, quality control, SOP development, experimental validation, system commissioning

Certified Mission Critical Operator (CMCO)

Conference papers

M. Tangri, K. Rahul, **K. Adsul**, M. Emami, Y. Fulpagare, D. Agonafer, "A comparative study of deep learning architectures for automated classification of particulate contaminants on electronic packaging surfaces," *ASME InterPACK 2026*

M. Tangri, **K. Adsul**, "Toward compute-efficient, boundary-condition-aware machine learning surrogates for tile-based IC thermal prediction," *IEEE ITherm 2026*

P. Jayaramu, M. Emami, V. Talari, M. R. Islam, **K. Adsul**, R. K. Suthar, et al., "Thermo-hydraulic performance of targeted flow in aluminum and copper heat sinks for immersion cooling applications," *IEEE ITherm 2025, Dallas, TX, pp. 1–9. doi: 10.1109/ITherm55376.2025.11235582*

M. Emami, R. K. Suthar, P. Bansode, M. Tangri, U. Rahangdale, S. Barua, D. Bhattarai, S. Amininasab, **K. K. Adsul**, D. Agonafer, "Investigation of the thermomechanical behavior of ASTRA M77 substrate core under single-phase liquid immersion cooling," *ASME InterPACK 2025, IPACK2025-169286. doi: 10.1115/IPACK2025-169286*

S. Saini, **K. K. Adsul**, P. Shahi, A. Niazmand, P. Bansode, D. Agonafer, "CFD modeling of the distribution of airborne particulate contaminants inside data center hardware," *ASME InterPACK 2020, IPACK2020-2590. doi: 10.1115/IPACK2020-2590*

Posters, thesis and manuscripts in preparation

POSTERS

K. Adsul, T. Lambe, U. Chinomso, R. Paranjpee, P. Satpute, T. Mistry, Y. Fulpagare, D. Agonafer, "Experimental characterization of nozzle spray distribution at varying heights on a crossflow heat exchanger," *InterPACK 2026*

K. Adsul, Y. Fulpagare, D. Agonafer, "Experimental mapping of spray distribution under variable nozzle height and operating conditions in a crossflow heat exchanger," *ES2 2026*

K. Adsul, S. Saini, P. Shahi, A. Niazmand, P. Bansode, D. Agonafer, "CFD modeling of the distribution of airborne particulate contaminants inside data center hardware," *ASME InterPACK 2020*

THESIS

K. K. Adsul, "Thermal design analysis of server chassis manifolds for liquid-cooled servers using CFD," *M.S. thesis, The University of Texas at Arlington, 2021*

IN PREPARATION

K. Adsul, L. Reddy, U. Chinomso, R. Paranjpee, P. Satpute, T. Mistry, Y. Fulpagare, D. Agonafer, "Experimental characterization of nozzle height on secondary-air heat transfer in a crossflow indirect evaporative cooling test facility under variable operating conditions," *journal manuscript in preparation*

K. Adsul, T. Lambe, T. Mistry, Y. Fulpagare, D. Agonafer, "Experimental mapping of collected-water distribution in a crossflow indirect evaporative cooling test facility under variable nozzle height, water flow, and fan speed," *journal manuscript in preparation*

LET'S CONNECT

Thank you

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